

BRAC UNIVERSITY
Department of Computer Science and Engineering

CSE 360: Computer Interfacing

Midterm Examination

Semester: Fall 2024

SET B

Duration: 1 Hour 10 minutes

Full Marks: 24

Answer All 3 following Questions.

[Figures in the right margin indicate marks]

CO1	Question 1. a) Differentiate between the internal and external interfacing of a computer with examples. b) Why is tri-state logic important in a multi-master and multi-slave interfacing? Explain the tri-state logic with appropriate diagrams and a truth table. c) What is a glitch? Identify the main cause of a glitch. d) Nowadays, the USB port is the most used connection point for data transfer all over the globe, why? Justify your answer with proper reasoning. e) Suppose there are 3 peripheral devices P1, P2 and P3 connected through buses with a bus master. P1 can only transmit data to the system bus, P2 can send and receive data to or from the bus master and P3 can only receive. i) Draw a bus configuration with all devices mentioned in the above scenario. ii) Identify what type of peripheral the P2 is.	[1] [3] [1] [1] [2]
CO2	Question 2. (i) Consider you are designing a single system with 3 STM32's with a couple of I/O devices that send continuous signals. a) Mention which communication protocol would be suitable for the scenario and why. b) Justify if it is possible to send data in full duplex mode using your chosen protocol. If not, then name the protocol that can send data in full duplex mode for continuous data transmission and how. [Hint: it cannot support multiple masters.] (ii)	[1.5] [2]

	In a different scenario, two devices X and Y use a different protocol that has an error-checking mechanism when sending data in spurts. a) Write the name of the communication protocol used in this case, and mention if it is possible to send data in full duplex mode using this protocol. [1.5] b) If X wants to send the data '10001110' to Y. Write the data packet the protocol would send from X to Y with proper <u>bit lengths</u> and <u>binary values</u> in your answer script. [3]	
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CO3	Question 3. (i) Suppose, you have an STM32 microcontroller and a device X. You have connected the device X with GPIO port C of the given STM32. You want to handle the <u>input signals using pin 5</u> and the <u>output signals using pin 9</u> of the same port. In this system, the <u>input signals are high (pull-up)</u> whereas <u>output signals are in reset state and output signals are sent at fast speed.</u> a) Draw the simplified architecture of STM32 microcontroller with appropriate labeling. [1] (b) Write all the necessary configuration commands for the given scenario. [3] Briefly describe all the registers that you have used to configure the system. (ii) A few commands are listed below: 1. $RCC \rightarrow AHB1EN = (1 << 2);$ 2. $RCC \rightarrow AHB1EN = (1 << 3);$ 3. $RCC \rightarrow AHB1EN = (1 << 5);$ 4. $RCC \rightarrow AHB1EN \&= (1 << 4);$ 5. $RCC \rightarrow AHB1EN = (1 << 2);$ a) Explain the operations of the CRC register. [1] b) Mention which GPIO ports will be activated after executing the above commands. Also, explain the reasoning behind it. [3]	
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